Angle Modulation

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Angle Modulation

Outlines

- Basics of Angle Modulation
- Advantages of Angle Modulation
- Applications of Angle Modulation

Basics of Angle Modulation

- In Angle Modulation, the Frequency or Phase of the Carrier Signal Changes with respect to the Modulating Signal.
- There are Two Types of Angle Modulation.
  - Frequency Modulation
  - Phase Modulation
- In Frequency Modulation, the Frequency of the Carrier Signal changes with respect to the Modulating Signal.

  - Let the Carrier Signal is given by:

    $\Longrightarrow \mathbf{C(t) = V_C \cos(\omega_c t + \phi)}$

  - Then, the Frequency Modulated Signal is given by:

    $\Longrightarrow \mathbf{Y_{FM}(t) = V_C \cos(\omega(t)\, t + \phi)}$

  - **Here, $\boldsymbol{\omega(t)}$ is a function of the Message Signal $\mathbf{m(t)}$.**

- In Phase Modulation, the Phase of the Carrier Signal changes with respect to the Modulating Signal.

  - Let the Carrier Signal is given by:

    $\Longrightarrow \mathbf{C(t) = V_C \cos(\omega_c t + \phi)}$

  - Then, the Phase Modulated Signal is given by:

    $\Longrightarrow \mathbf{Y_{PM}(t) = V_C \cos(\omega_c t + \phi(t))}$

  - **Here, $\boldsymbol{\phi(t)}$ is a function of the Message Signal $\mathbf{m(t)}$.**

Advantages of Angle Modulation

1. **Improved noise immunity** compared to amplitude modulation.

- Most noise in communication channels affects the **amplitude** of signals.
- In **angle modulation**, **the amplitude of the carrier does not contain information**. The information is stored in:
  - Frequency (FM)
  - Phase (PM)
- So even if noise changes the amplitude, the information remains unaffected.

2. **Constant amplitude** reduces power loss due to non-linearities in amplifiers.

- In **angle modulation**, the carrier amplitude is **constant**.
- Because the amplitude does not change:
  - Amplifiers can operate in **non-linear modes (reduce power loss)**
  - High efficiency amplifiers can be used
- In contrast, in **AM** the amplitude varies.
- If a nonlinear amplifier is used with AM, **distortion occurs** because the amplitude contains information.

Advantages of Angle Modulation

3. **Better signal quality,** especially for FM radio (less distortion).

- **[Capture Effect]** If two FM signals arrive at the receiver:

  - **The stronger signal (determined by the received signal power) dominates**

  - The weaker signal is suppressed

- **AM** receivers mix both signals, **causing distortion**.

4. **Higher efficiency** in bandwidth use with proper techniques (e.g., narrowband FM).

- Narrowband FM reduces bandwidth significantly.

5. **Less susceptible to signal fading** and amplitude variations.

- Fading occurs when signal amplitude fluctuates due to:

  - multipath propagation
  - obstacles
  - atmospheric conditions

- In AM, fading directly affects the message

Applications of Angle Modulation

- **FM radio broadcasting** (88 – 108 MHz): High fidelity audio transmission.
- **Two-way radio communications** (police, fire, emergency services).
- **Television sound transmission** (Analog TV).
- **Satellite and space communication**: Used due to robustness against noise.
- **Radar and telemetry systems**.

Instantaneous Angular Frequency

- Instantaneous angular frequency is the rate of change of the instantaneous phase (angle) of a signal with respect to time.
- In other words, it tells us how fast the phase of a signal is changing at a particular instant. It represents the instantaneous rotational speed of the signal's phasor in radians per second.

Consider a general sinusoidal signal:

$$s(t) = A(t)\cos[\theta(t)]$$

where

- $A(t)$ = instantaneous amplitude
- $\theta(t)$ = instantaneous phase (angle)

Formula of Instantaneous Angular Frequency

The instantaneous angular frequency is defined as

$$\boxed{\omega_i(t) = \frac{d\theta(t)}{dt}}$$

where

- $\omega_i(t)$ = instantaneous angular frequency (rad/s)
- $\theta(t)$ = instantaneous phase (radians)

General Carrier Signal

A carrier signal is

$$y_c(t) = A_c \cos(\omega_c t)$$

where

- A_c = carrier amplitude
- ω_c = carrier angular frequency
- t = time

A general angle-modulated signal can be written as

$$y(t) = A_c \cos[\theta(t)]$$

where

$$\theta(t) = \text{instantaneous phase}$$

Derivation of Phase Modulation (PM)

Step 1: Definition of PM

Phase Modulation:

The **instantaneous phase of the carrier varies linearly with the message signal**.

$$\theta(t) = \omega_c t + k_p m(t)$$

where

- $m(t)$ = message signal
- k_p = phase sensitivity constant

Step 2: Substitute Instantaneous Phase into Signal

The general signal equation:

$$y(t) = A_c \cos[\theta(t)]$$

Substitute $\theta(t)$:

$$y_{PM}(t) = A_c \cos[\omega_c t + k_p m(t)]$$

Instantaneous angular frequency is defined as

$$\omega_i(t) = \frac{d\theta(t)}{dt}$$

i.e., **the derivative of instantaneous phase with respect to time**.

Substitute the phase expression:

$$\omega_i(t) = \frac{d}{dt}[\omega_c t + k_p m(t)]$$

Differentiate term by term:

$$\omega_i(t) = \frac{d(\omega_c t)}{dt} + k_p \frac{dm(t)}{dt}$$

$$\omega_i(t) = \omega_c + k_p \frac{dm(t)}{dt}$$

Derivation of Frequency Modulation (FM)

Step 1: Definition of FM

Frequency Modulation:

The **instantaneous frequency varies linearly with the message signal.**

$$\omega_i(t) = \omega_c + k_f m(t)$$

where

- $\omega_i(t)$ = instantaneous angular frequency
- k_f = frequency sensitivity (k_f has units of rad/(s·V).)

Step 2: Relation Between Frequency and Phase

Instantaneous angular frequency is

$$\omega_i(t) = \frac{d\theta(t)}{dt}$$

Substitute the FM definition:

$$\frac{d\theta(t)}{dt} = \omega_c + k_f m(t)$$

Step 3: Integrate Both Sides

$$\theta(t) = \int [\omega_c + k_f m(t)] dt$$

$$\theta(t) = \omega_c t + k_f \int m(t) dt$$

Step 4: Substitute into General Signal

$$y(t) = A_c \cos[\theta(t)]$$

$$y_{FM}(t) = A_c \cos\left[\omega_c t + k_f \int m(t) dt\right]$$

Important Concept

$$FM = PM \text{ of } \int m(t)dt$$

$$PM = FM \text{ of } \frac{dm(t)}{dt}$$

Frequency Modulation

Outlines

- Basics of Frequency Modulation
- Block Diagram of Frequency Modulator
- Waveforms of Frequency Modulator
- Square Wave Frequency Modulation

Basics of Frequency Modulation

- In Frequency Modulation, the Frequency of the Carrier Signal changes with respect to the Modulating Signal.

  - Let the Carrier Signal is given by:

    $$\Longrightarrow \mathbf{C(t)} = \mathbf{V_C \cos(\omega_c t)} = \mathbf{V_C \cos(\theta(t))}$$

  - Then, the Frequency Modulated Signal is given by:

    $$\Longrightarrow \mathbf{Y_{FM}(t)} = \mathbf{V_C \cos(\omega(t)\, t)}$$

  - Here, $\boldsymbol{\omega}(\mathbf{t})$ and $\boldsymbol{\theta}(\mathbf{t})$ is a function of the Message Signal $\mathbf{m(t)}$.

- Frequency Modulation is a part of the Angle Modulation, The Angle of Frequency Modulation is given by:

  $$\Longrightarrow \boldsymbol{\theta}(\mathbf{t}) = \boldsymbol{\omega_c}\mathbf{t} + 2\pi \mathbf{k_F} \int_0^t m(\alpha)\, d\alpha$$

- $\mathbf{k_F}$ is frequency Sensitivity, Unit is Hz/Volt.

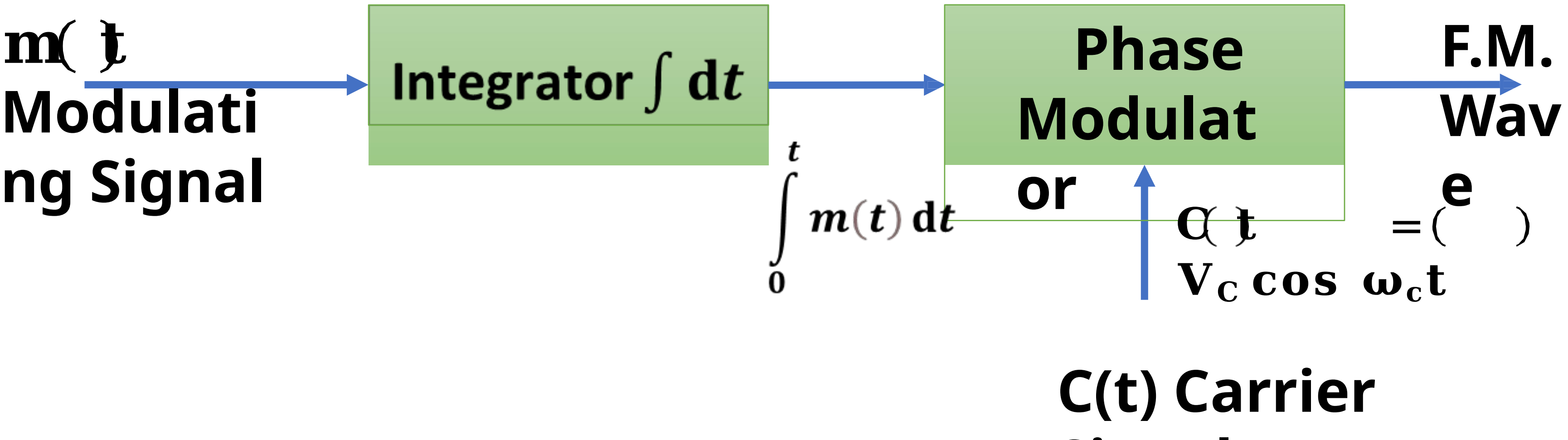
Block Diagram of Frequency Modulator

- First, The Message Signal $m(t)$ is given to the Integrator.
- Then the Integrated Message Signal and Carrier Signal is given to the Phase Modulator.
- Phase Modulator generates a Frequency Modulated wave and its Angle is given by:

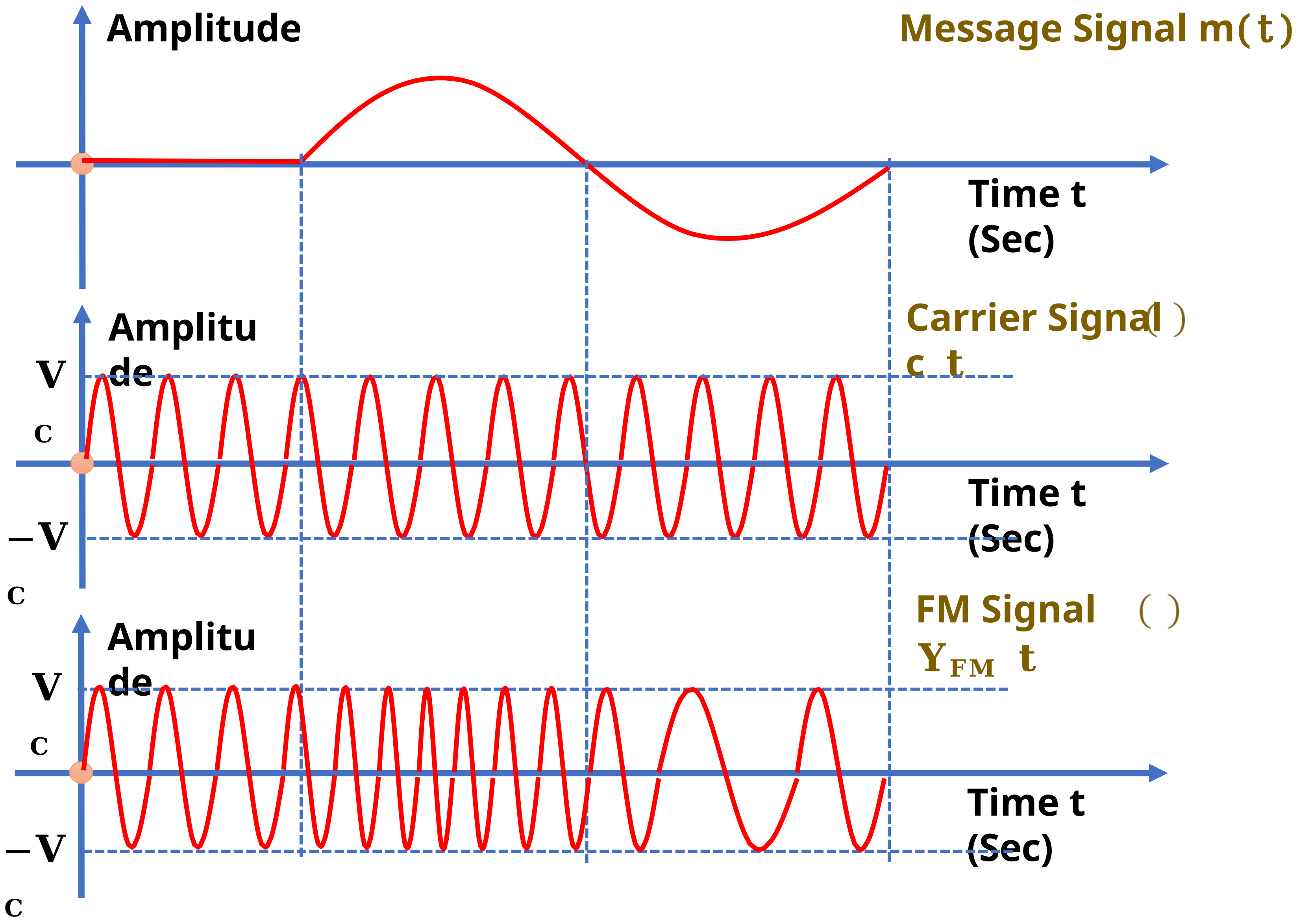
$$\Longrightarrow \theta(t) = \omega_c t + 2\pi k_F \int_0^t m(t)\,dt$$

- Here, k_F is frequency Sensitivity, Unit is Hz/Volt.
- Frequency Modulated Wave is given by:

$$\Longrightarrow Y_{FM}(t) = V_C \cos\left(\omega_c t + 2\pi k_F \int_0^t m(t)\,dt\right)$$



Waveforms of Frequency Modulator

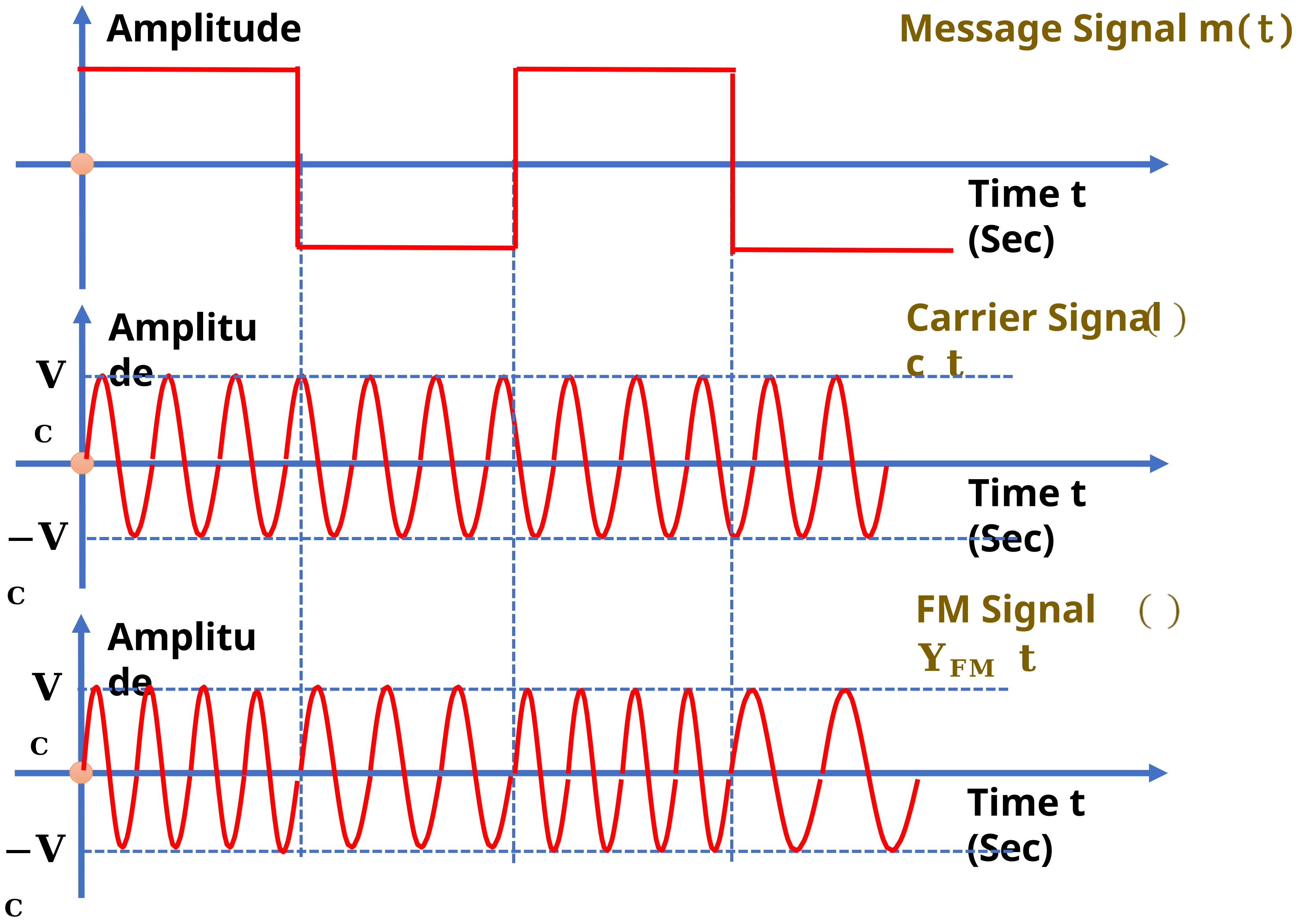


$$\omega_i(t) = \omega_c + k_f m(t)$$
$$k_f\ unit \Rightarrow rad/(sec.Volt)$$

$$f_i(t) = f_c + k_f m(t)$$
$$k_f\ unit \Rightarrow Hz/Volt$$

$$2\pi f_i(t) = 2\pi f_c + 2\pi k_f m(t)$$
$$\Rightarrow \omega_i(t) = \omega_c + 2\pi k_f m(t)$$
$$k_f\ unit \Rightarrow Hz/Volt$$

Square Wave Frequency Modulation



$$\boxed{\omega_i(t) = \omega_c + k_f m(t)}$$

$$\boxed{f_i(t) = f_c + k_f m(t)}$$

$$\boxed{\omega_i(t) = \omega_c + 2\pi k_f m(t)}$$

Phase Modulation

Outlines

- Basics of Phase Modulation
- Block Diagram of Phase Modulator
- Waveforms of Phase Modulator
- Square Wave Phase Modulator

Basics of Phase Modulation

- **In Phase Modulation, the Phase of the Carrier Signal changes with respect to the Modulating Signal.**

  - Let the Carrier Signal is given by:

    $$\Longrightarrow C(t) = V_C \cos(\omega_c t + \phi) = V_C \cos\theta(t)$$

  - Then, the Phase Modulated Signal is given by:

    $$\Longrightarrow Y_{PM}(t) = V_C \cos(\omega_c t\;)$$

  - **Here, ϕ and $\theta(t)$ is a function of the Message Signal $m(t)$.**

- **Phase Modulation is a part of the Angle Modulation, The Angle of Phase Modulation is given by:**

  $$\Longrightarrow \theta(t) = \omega_c t + k_P m(t)$$

- **k_P is Phase Sensitivity, Unit is Rad/Volt.**

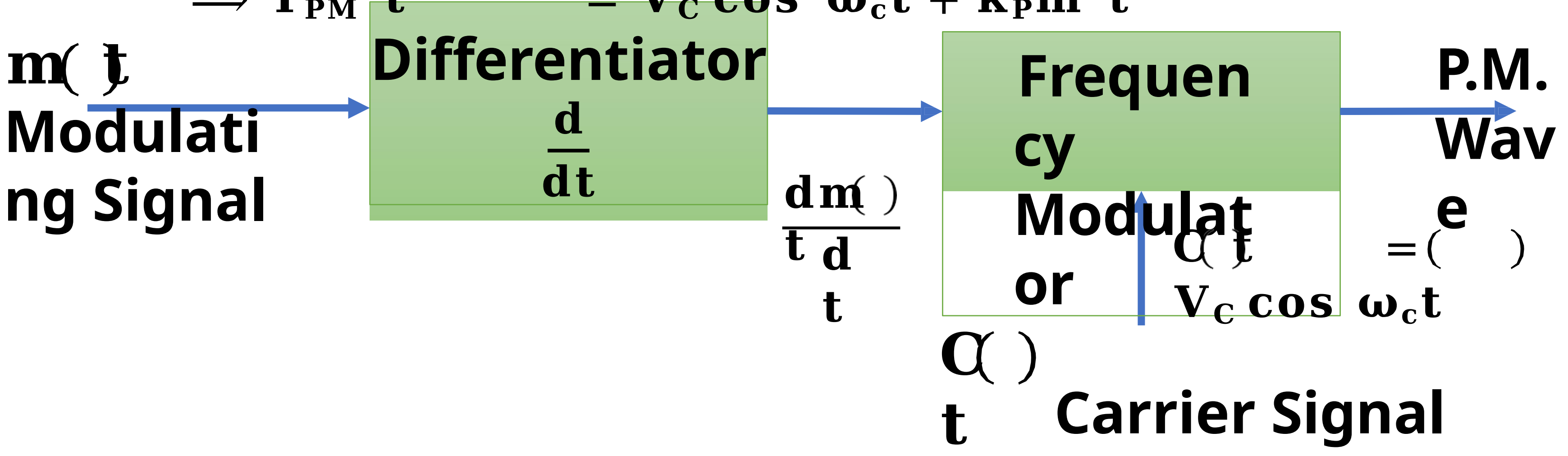
Block Diagram of Phase Modulator

- First, the Message Signal $m(t)$ is given to the Differentiator.
- Then the Differentiated Message Signal and the Carrier Signal are given to the Frequency Modulator.
- Frequency Modulator generates Phase Modulated wave and its Angle is given by:

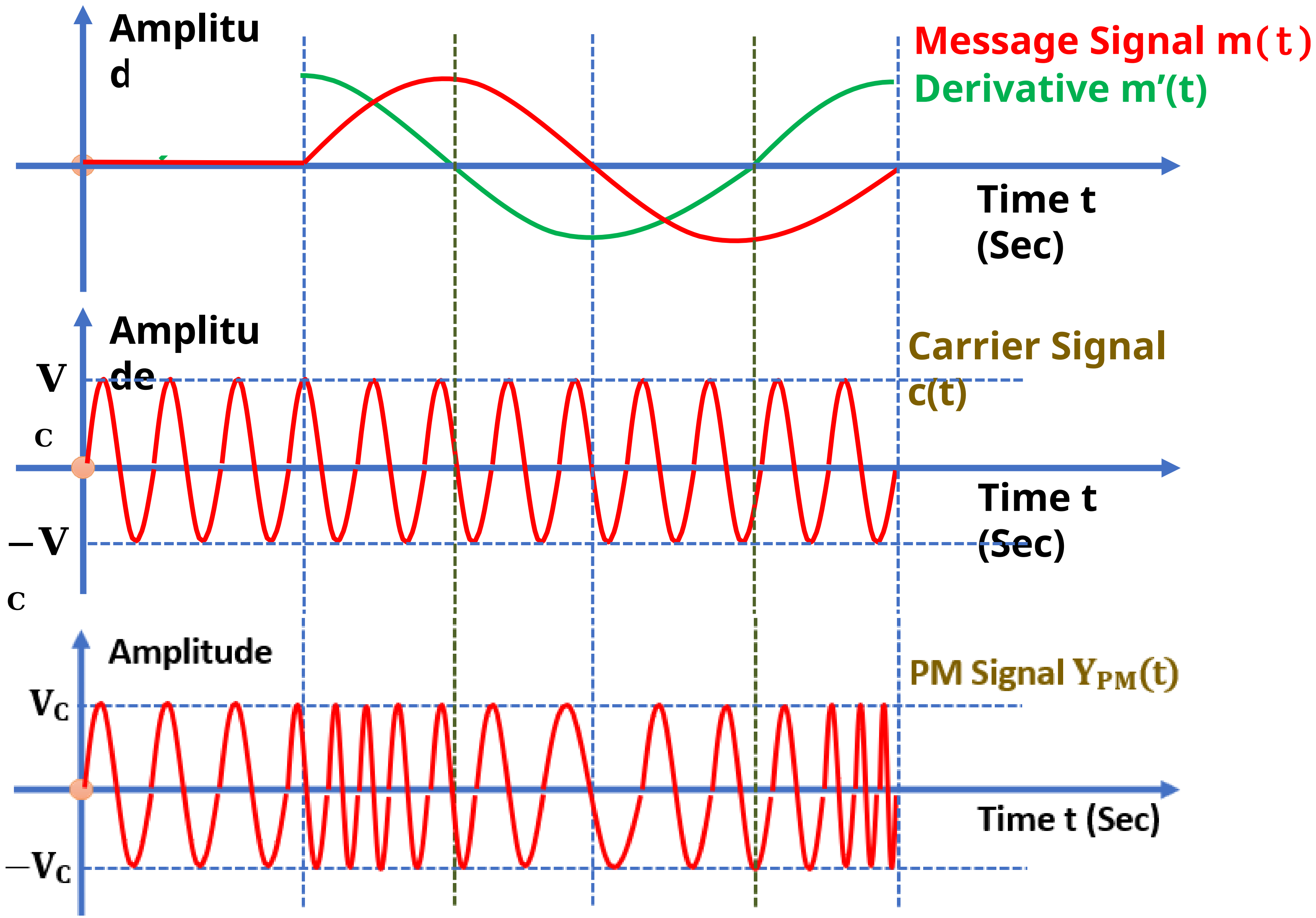
$$\Rightarrow \theta(t) = \omega_c t + k_P m(t)$$

- Here, k_P is Phase Sensitivity, Unit is Rad/Volt.
- Phase Modulated Wave is given by:

$$\Rightarrow Y_{PM}(t) = V_C \cos(\omega_c t + k_P m(t))$$



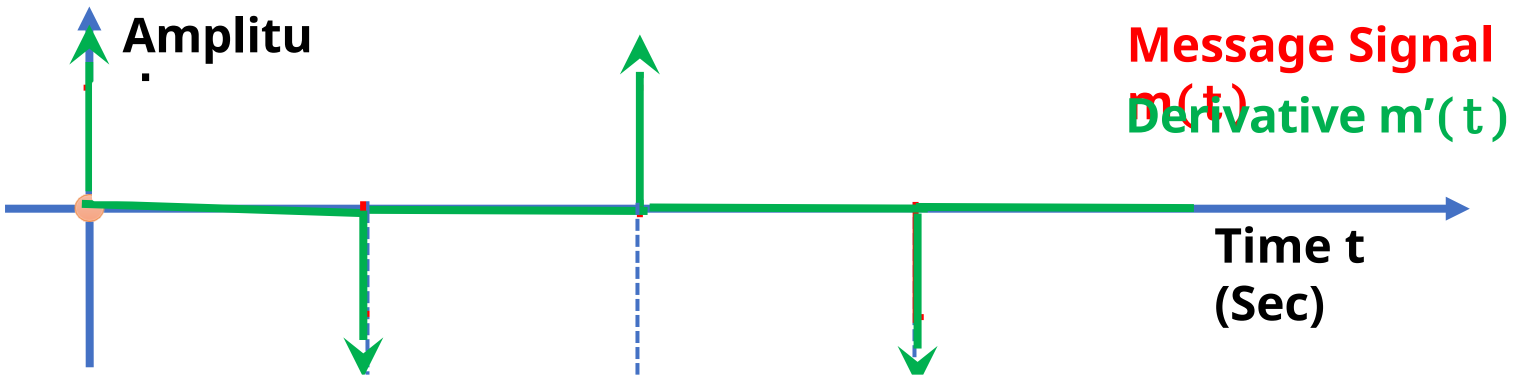
Waveforms of Phase Modulator



$$\theta(t) = \omega_c t + k_p m(t)$$

$$\frac{d\theta(t)}{dx} = \omega_i(t) = \omega_c + k_p m'(t)$$

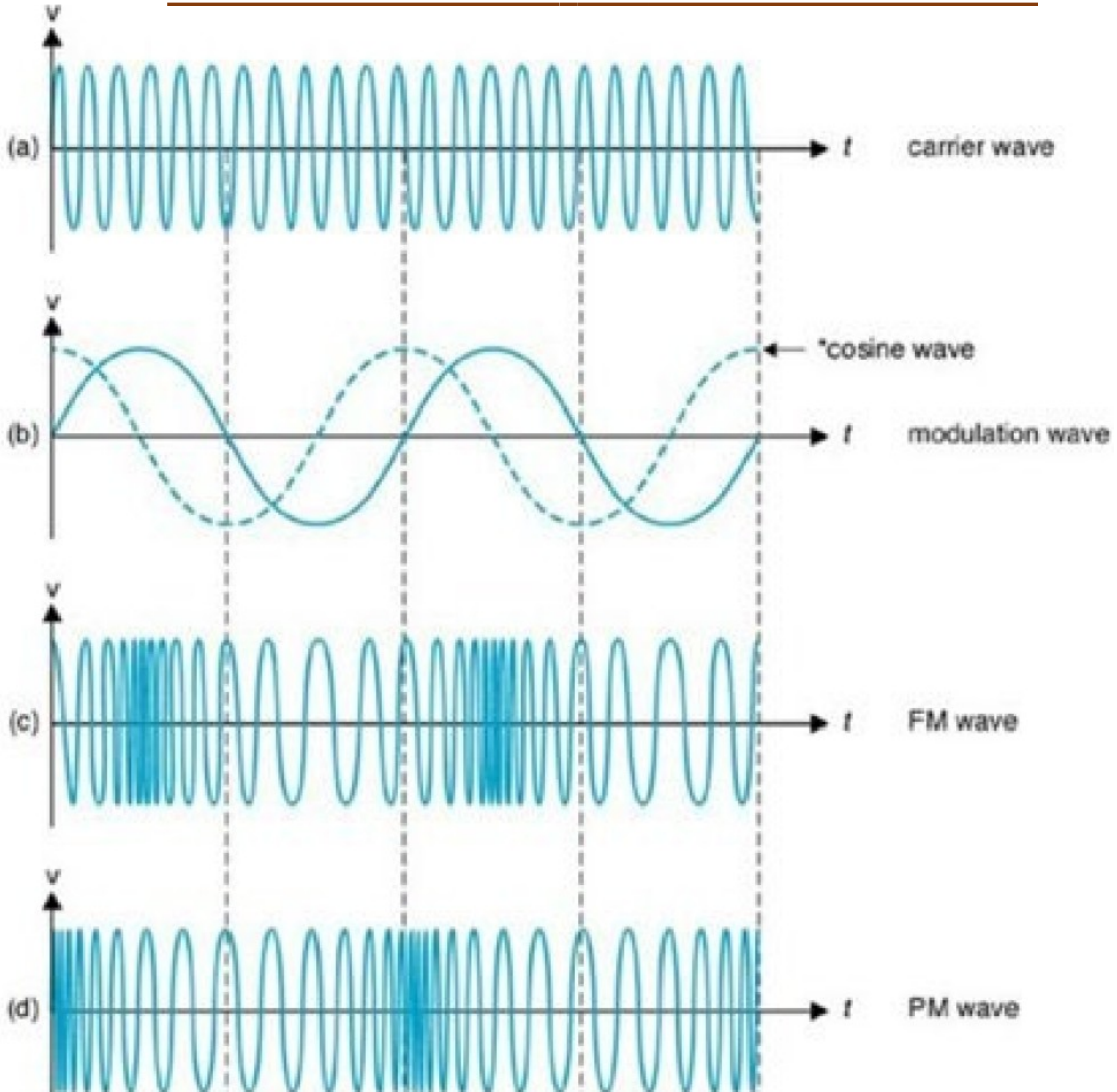
Square Wave Phase Modulation



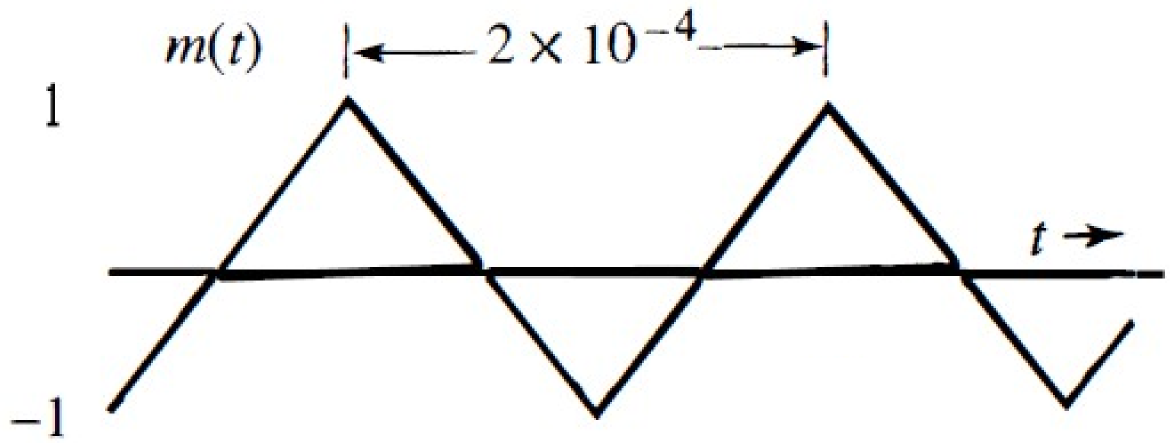
$$\theta(t) = \omega_c t + k_p m(t)$$

$$\frac{d\theta(t)}{dx} = \omega_i(t) = \omega_c + k_p m'(t)$$

Waveforms of FM and PM

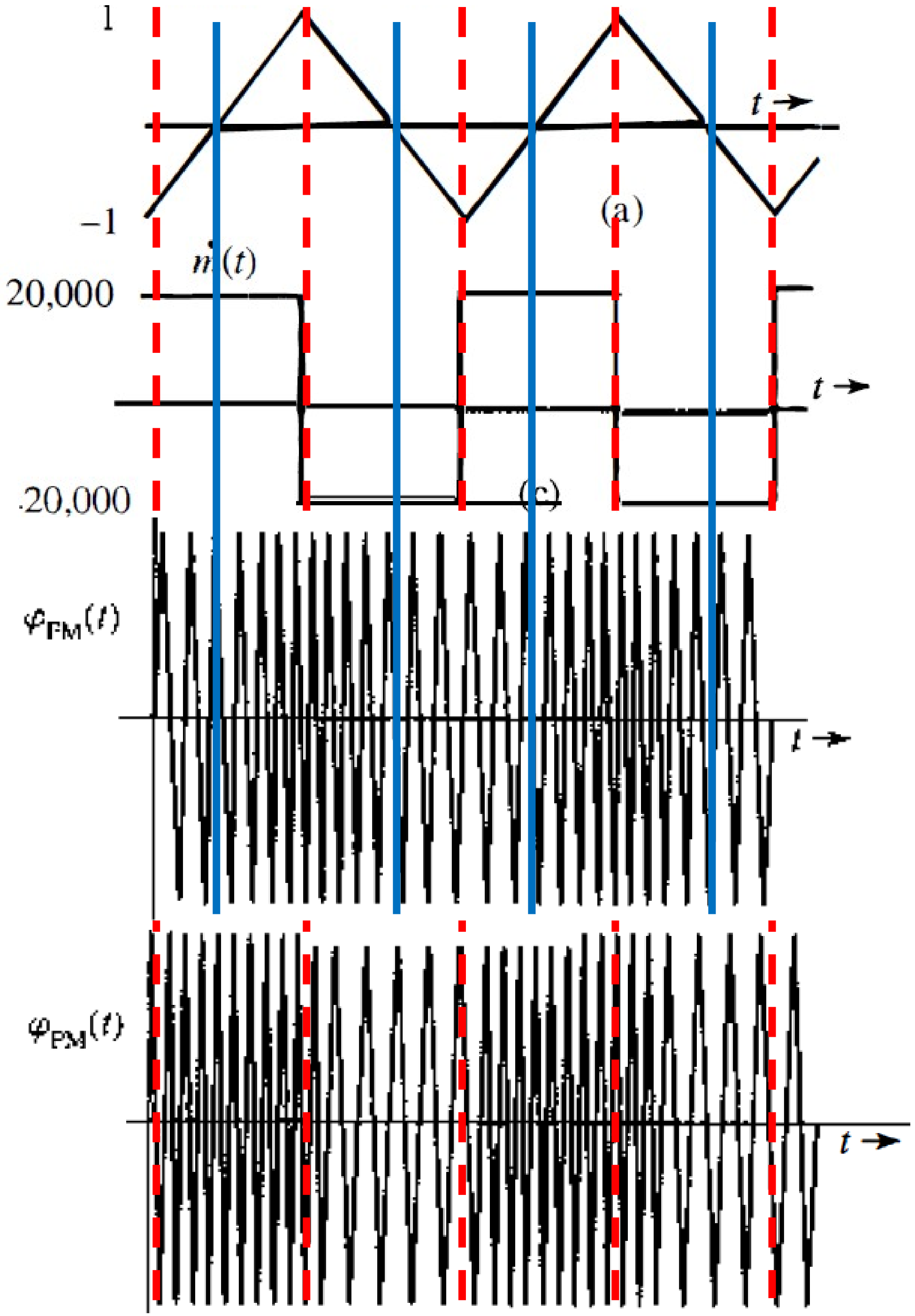


Problem 1. Sketch FM and PM waves for the modulating signal ***m(t)*** shown in Figure below. The constants k_f and k_p are $2\pi \times 10^5\, rad/(sec.Volt)$ and $10\pi\, rad/Volt$ respectively, and the carrier frequency f_c is $100\ MHz$. [**Lathi, Zhi Ding: Example 5.1**]



Solve:

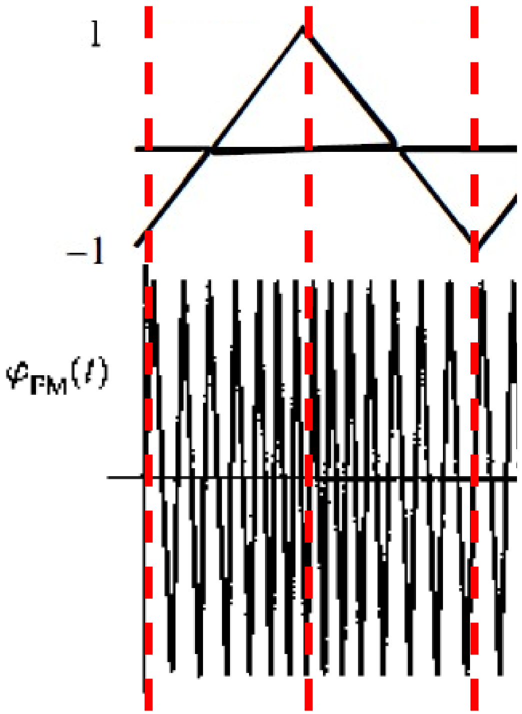
Waveform



For FM:

$$\omega_i = \omega_c + k_f m(t)$$

Dividing throughout by 2π, we have the equation in terms of the variable f (frequency in hertz). The instantaneous frequency f_i is



$$f_i = f_c + \frac{k_f}{2\pi} m(t)$$

$$= 10^8 + 10^5 m(t)$$

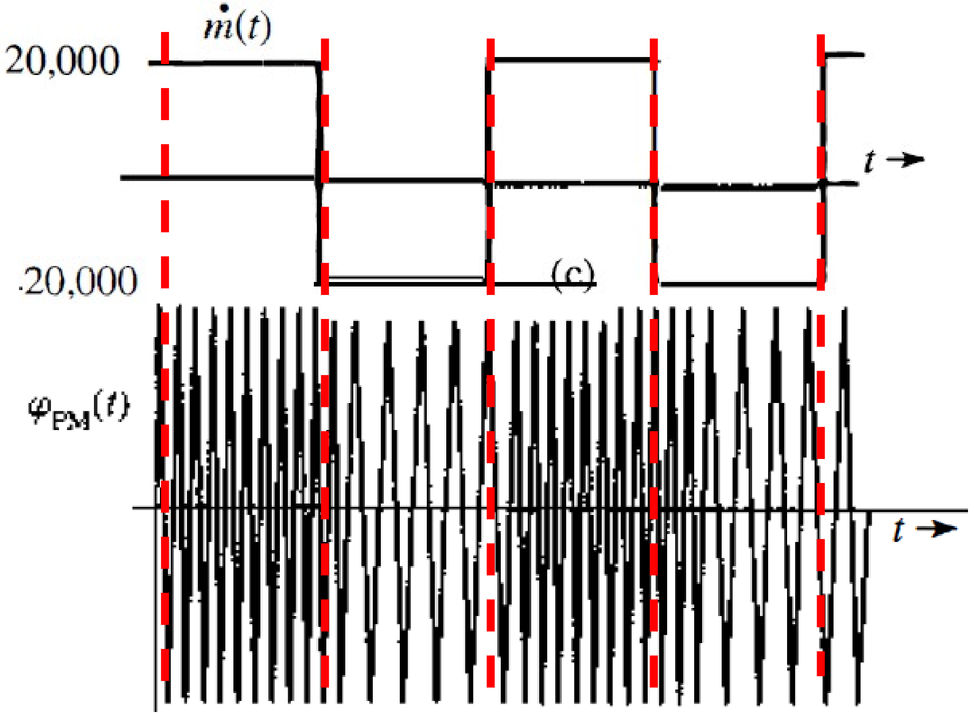
$$(f_i)_{\min} = 10^8 + 10^5\,[m(t)]_{\min} = 99.9 \text{ MHz}$$

$$(f_i)_{\max} = 10^8 + 10^5\,[m(t)]_{\max} = 100.1 \text{ MHz}$$

Because $m(t)$ increases and decreases linearly with time, the instantaneous frequency increases linearly from 99.9 to 100.1 MHz over a half-cycle and decreases linearly from 100.1 to 99.9 MHz over the remaining half-cycle of the modulating signal

PM for $m(t)$ is FM for $\dot{m}(t)$.

For PM:



$$f_i = f_c + \frac{k_p}{2\pi}\dot{m}(t)$$

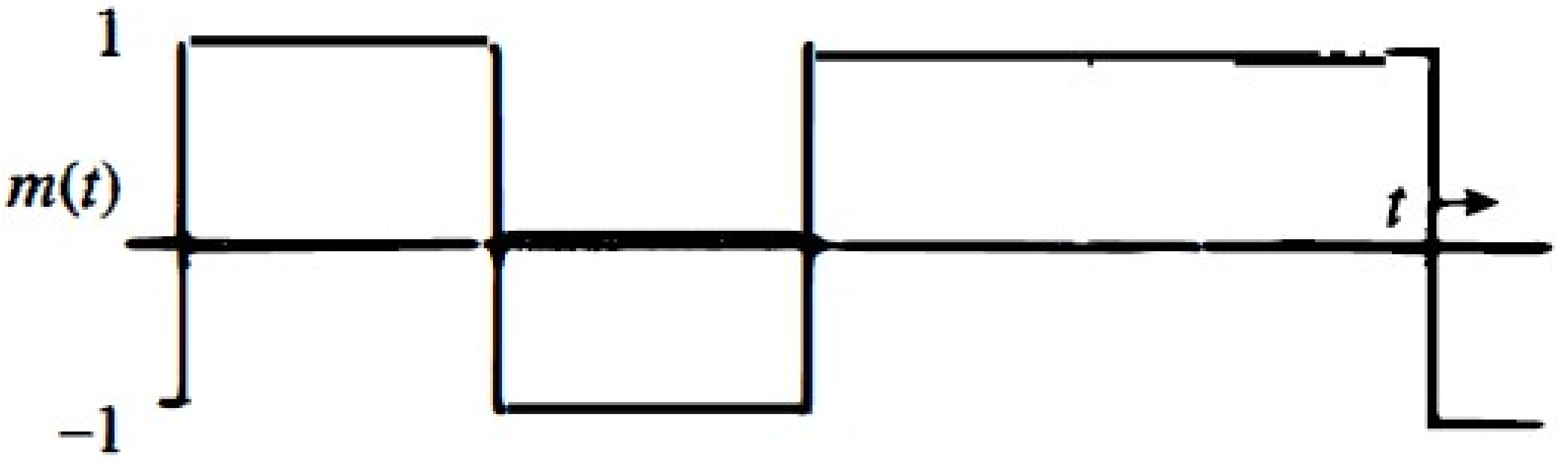
$$= 10^8 + 5\,\dot{m}(t)$$

$$(f_i)_{\min} = 10^8 + 5\,[\dot{m}(t)]_{\min} = 10^8 - 10^5 = 99.9 \text{ MHz}$$

$$(f_i)_{\max} = 10^8 + 5\,[\dot{m}(t)]_{\max} = 100.1 \text{ MHz}$$

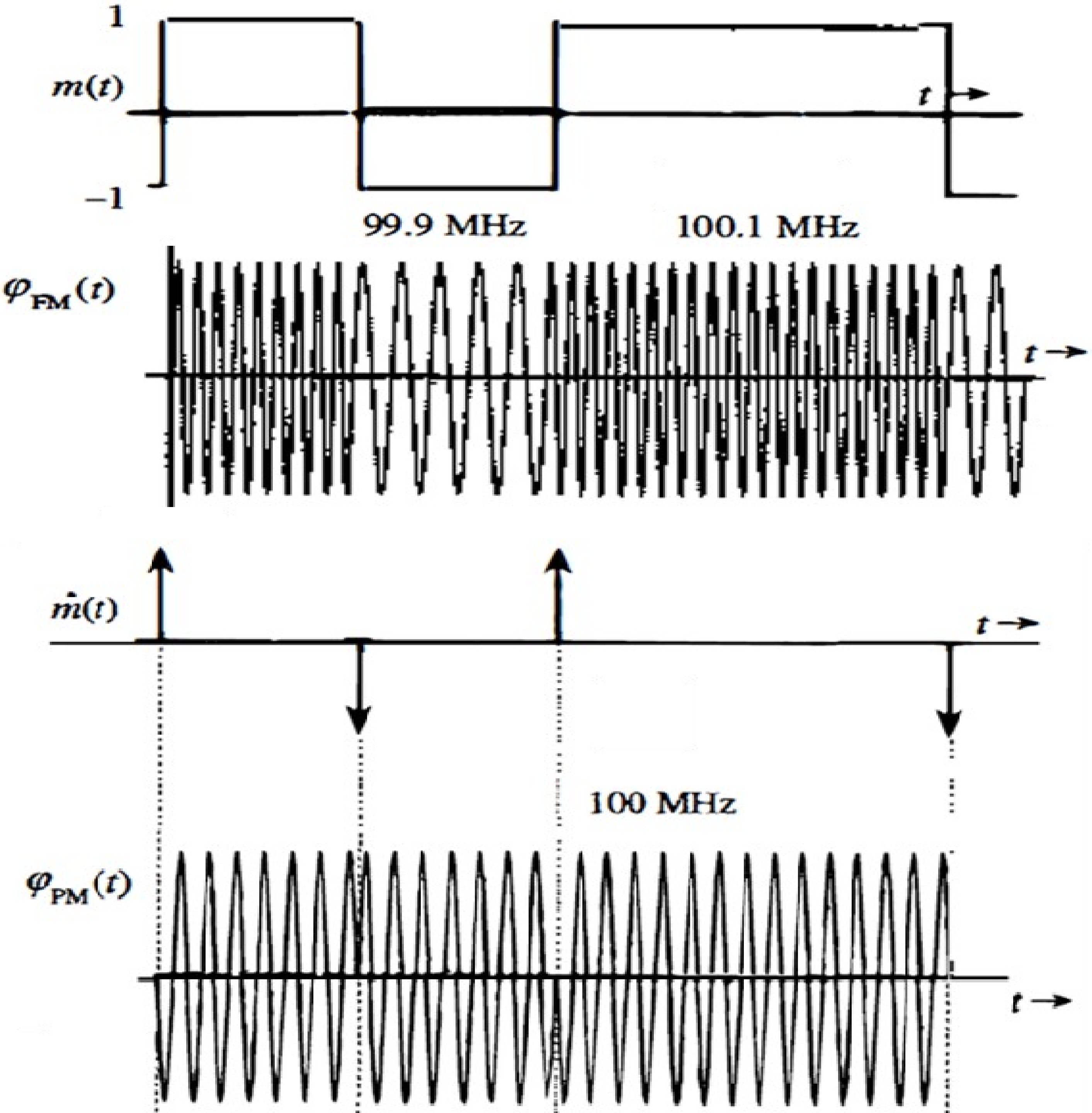
Because $\dot{m}(t)$ switches back and forth from a value of $-20{,}000$ to 20,000, the carrier frequency switches back and forth from 99.9 to 100.1 MHz every half-cycle of $\dot{m}(t)$, as shown in Fig.

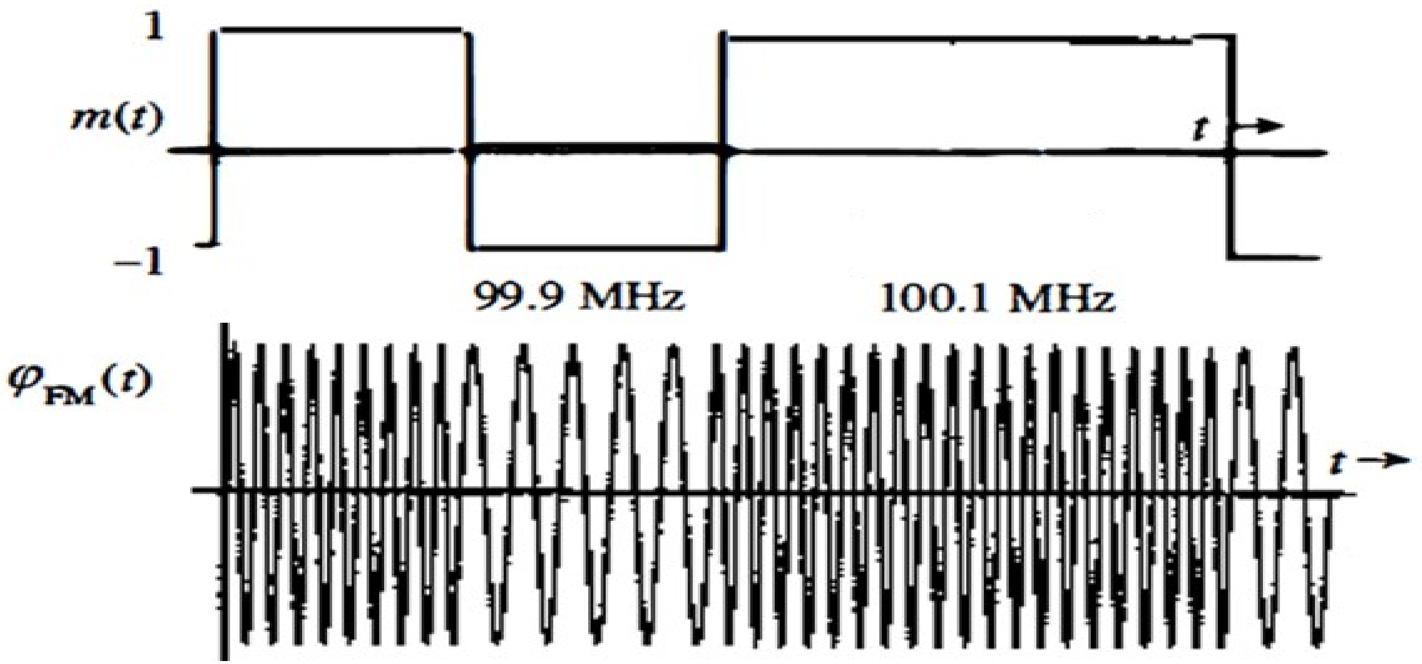
Problem 2. Sketch FM and PM waves for the modulating signal ***m(t)*** shown in the figure below. The constants k_f and k_p are $2\pi \times 10^5\ rad/(sec.Volt)$ and $\frac{\pi}{2}\ rad/Volt$ respectively, and the carrier frequency f_c is $100\ MHz$. Consider the carrier signal $A cos(\omega_c t)$ [**Lathi, Zhi Ding: Example 5.2**]



Solve:

Waveform

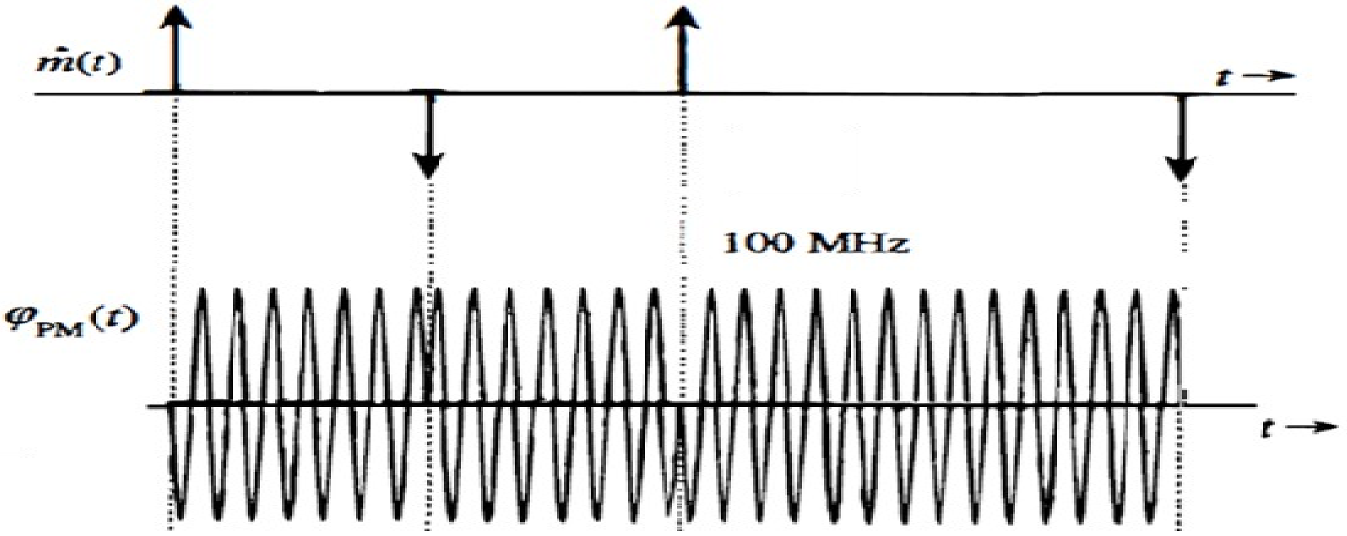




For FM:

$$f_i = f_c + \frac{k_f}{2\pi} m(t) = 10^8 + 10^5 m(t)$$

Because $m(t)$ switches from 1 to -1 and vice versa, the FM wave frequency switches back and forth between 99.9 and 100.1 MHz, as shown in Fig.



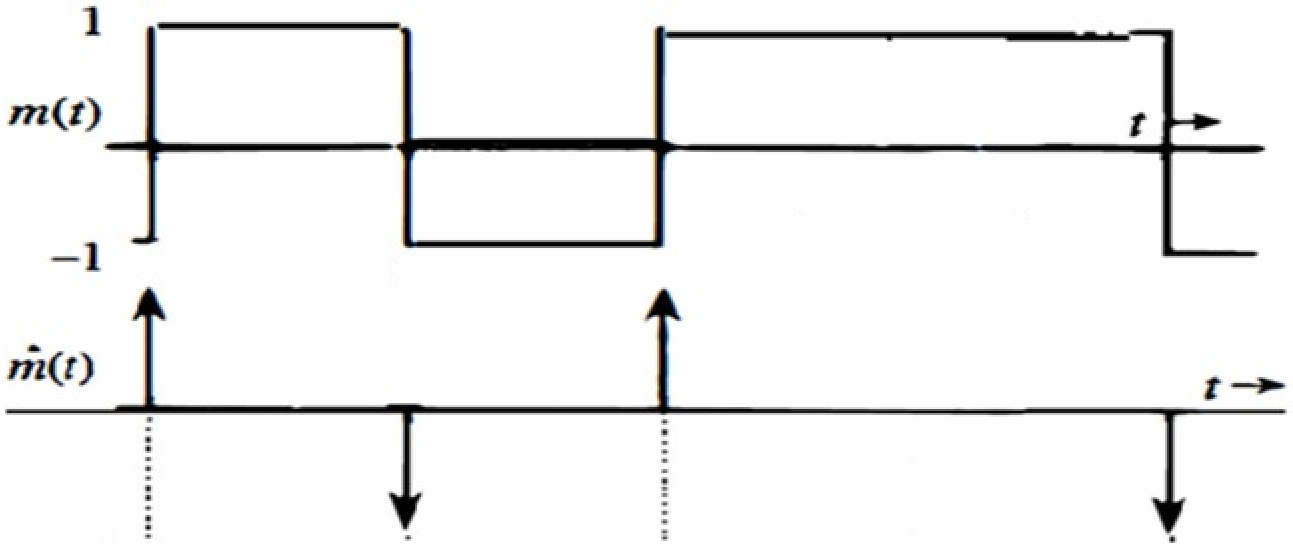
For PM:

$$f_i = f_c + \frac{k_p}{2\pi}\dot{m}(t) = 10^8 + \frac{1}{4}\dot{m}(t)$$

The derivative $\dot{m}(t)$ is zero except at points of discontinuity of $m(t)$ where impulses of strength ± 2 are present. This means that the frequency of the PM signal stays the same except at these isolated points of time! It is not immediately apparent how an instantaneous frequency can be changed by an infinite amount and then changed back to the original frequency in zero time. Let us consider the direct approach:

$$\begin{aligned}\varphi_{PM}(t) &= A\cos\left[\omega_c t + k_p m(t)\right]\\ &= A\cos\left[\omega_c t + \frac{\pi}{2}m(t)\right]\\ &= \begin{cases} A\sin\omega_c t & \text{when } m(t) = -1\\ -A\sin\omega_c t & \text{when } m(t) = 1\end{cases}\end{aligned}$$

Phase Discontinuity and Phase Shift



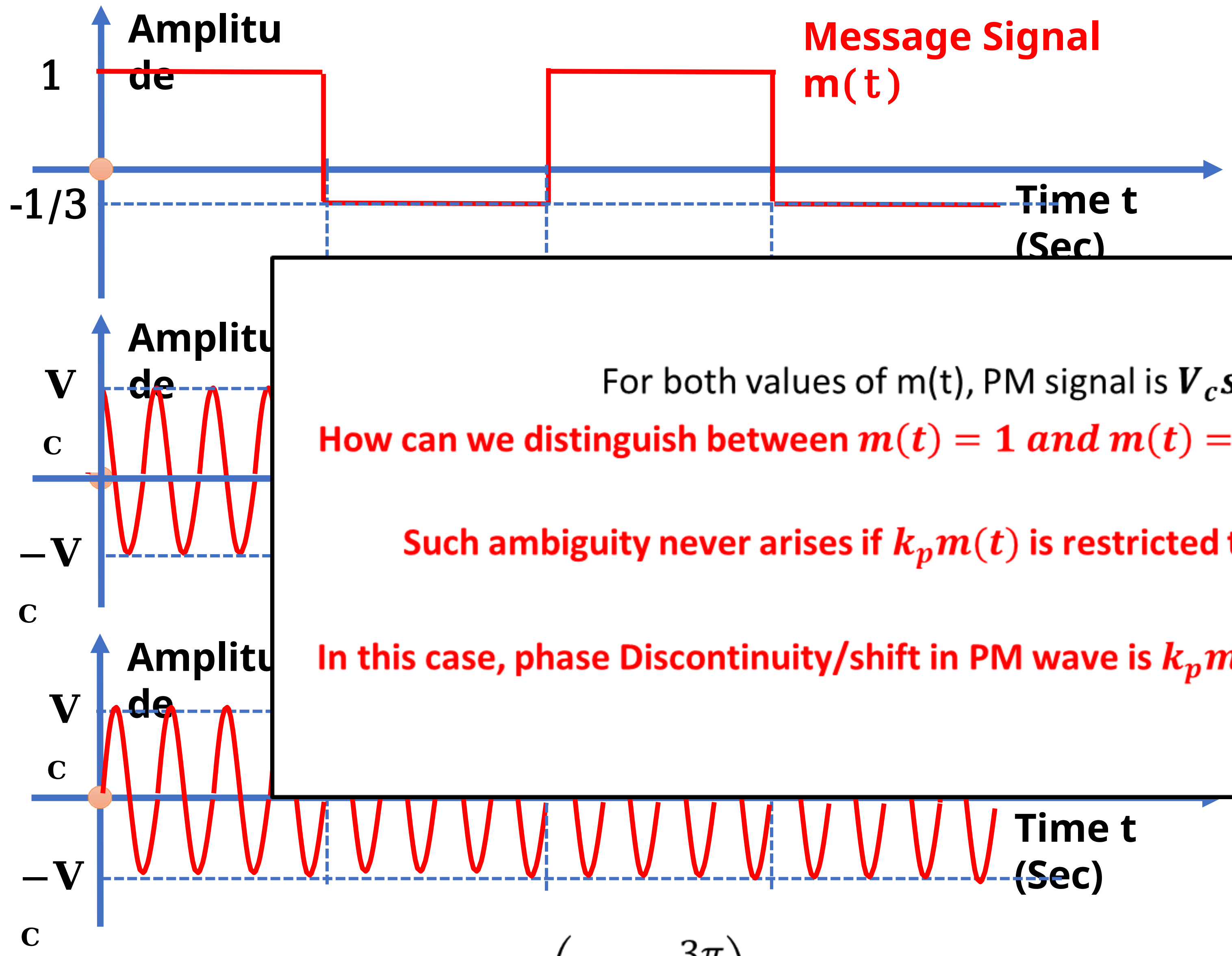
$$\varphi_{PM}(t) = A \cos[\omega_c t + k_p m(t)]$$

- The PM wave $\phi_{PM}(t)$ in this case has phase discontinuities at instants where impulses of $m'(t)$ are located. At these instants, the carrier phase shifts by $\boldsymbol{\pi}$ instantaneously.
- A finite phase shift in zero time implies infinite instantaneous frequency at these instants.
- **Phase discontinuity/shift measurement**: The amount of phase discontinuity in $\phi_{PM}(t)$ at the instant where $m(t)$ is discontinuous is $k_p m_d$, where m_d is the amount of discontinuity in $m(t)$ at that instant.
- For the previous example,

  - $m_d = 2$ $[the\ amplitude\ of\ m(t) changes\ by\ 2\ (from\ \text{-}1\ to\ 1) at\ the\ discontinuity]$

  - $the\ phase\ discontinuity\ in\ \phi_{PM}(t)\ is\ k_p m_d = \frac{\pi}{2} \times 2 = \pi\ rad$

Ambiguity in Demodulation



For both values of m(t), PM signal is $\boldsymbol{V_c \sin(\omega_c t)}$

How can we distinguish between $m(t) = 1$ *and* $m(t) = -\frac{1}{3}$ during demodulation?

Such ambiguity never arises if $k_p m(t)$ is restricted to the range $(-\pi, \pi)$.

In this case, phase Discontinuity/shift in PM wave is $k_p m_d = \left(\frac{3\pi}{2} \times \left(1 + \frac{1}{3}\right)\right) = 2\pi$

$$\ldots_c t + k_p m(t)\big)$$

$$\Rightarrow Y_{PM}(t) = V_c \cos\left(\omega_c t + \frac{3\pi}{2} m(t)\right)$$

$when\, m(t) = 1, Y_{PM}(t) = V_c \cos\left(\omega_c t + \frac{3\pi}{2}\right) = \boldsymbol{V_c \sin(\omega_c t)}$ $\quad$ $when\, m(t) = -\frac{1}{3}, Y_{PM}(t) = V_c \cos\left(\omega_c t - \frac{\pi}{2}\right) = \boldsymbol{V_c \sin(\omega_c t)}$

Block Diagrams of FM & PM Wave

Not a generalized block diagram!!!

Carrier Signal

Generalized Angle Modulation: Concept

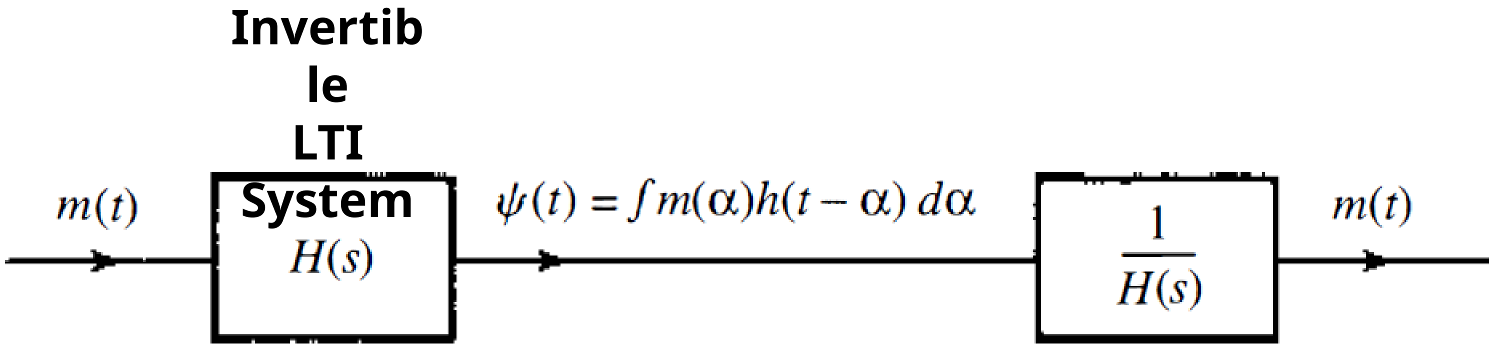
- **The General Equation:** The generalized angle-modulated signal is defined as:

$$\phi_{EM}(t) = A\cos[\omega_c t + \psi(t)]$$

- **Defining $\boldsymbol{\psi(t)}$:** The term $\psi(t)$ represents the processed message. It is obtained by convolving the message $m(t)$ with an impulse response $h(t)$:

$$\psi(t) = \int_0^t m(\alpha)h(t-\alpha)d\alpha$$

- **Interpretation:** This integral means that the phase of the carrier is being altered by a "filtered" version of the original message, rather than just the raw message itself.



Generalized Angle Modulation: System Architecture

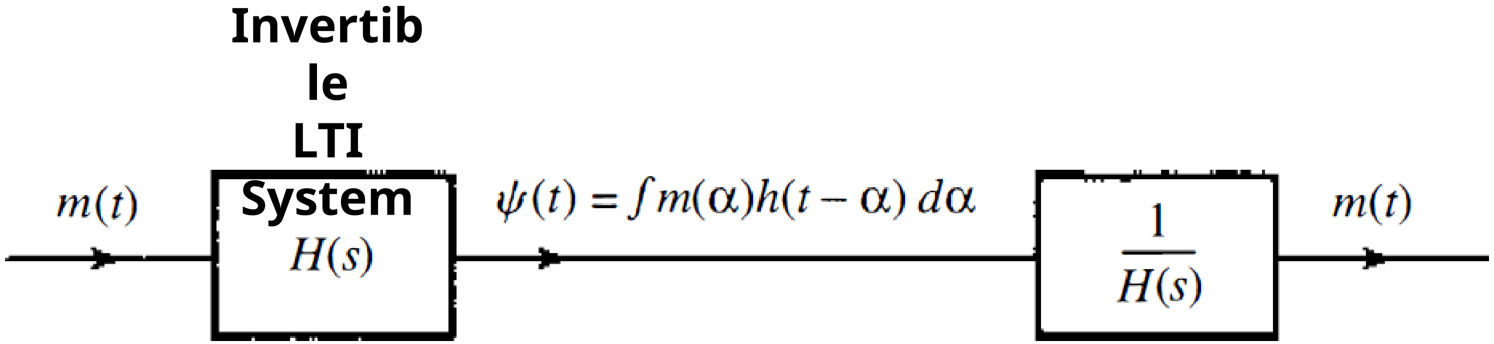
The Modulation Process (Transmitter)

- **Input Signal:** The process starts with the original message signal, $m(t)$.
- The Pre-processor ($H(s)$): The signal passes through a linear, time-invariant system with transfer function $H(s)$. This block transforms $m(t)$ into $\psi(t)$.
- **The Modulator**: The processed signal $\psi(t)$ is then used to modulate the angle of the carrier, producing the output $\phi_{EM}(t)$.

The Demodulation Process (Receiver)

- **Recovery Requirement:** To get the original message back, we must reverse the processing done at the transmitter.
- **The Inverse System** ($\frac{1}{H(s)}$): The received signal (or the recovered phase) is passed through a system with the inverse transfer function, $[H(s)]^{-1}$.
- **Output**: This successfully recovers the original message $m(t)$.
- **Crucial Constraint**: This recovery is only possible if $H(s)$ is an **invertible** operation (meaning no information was lost during the initial processing).

Generalized Angle Modulation: FM and PM as Special Cases



- **Phase Modulation (PM):** This is just a special case where the filter $H(s)$ acts as a simple scaler.

$$\textbf{Impulse Response: } h(t) = k_p \delta(t)$$

$$\textbf{Transfer function: } H(s) = k_p$$

Effect: The phase is directly proportional to $m(t) \Rightarrow \psi(t) = k_p m(t)$

- **Frequency Modulation (FM):** This is a special case where the filter acts as an integrator.

$$\textbf{Impulse Response: } h(t) = k_f u(t)$$

$$\textbf{Transfer function: } H(s) = \frac{k_f}{s}$$

Effect: The phase is proportional to the integral of $m(t) \Rightarrow \psi(t) = k_f \int_0^t m(\alpha)d\alpha$

Why PM is Practically Superior?

- **No DC Drift:** FM's integration can cause carrier frequency drift
- **Better Phase Stability:** PM maintains better phase relationships
- **Easier Synchronization:** PM systems often have cleaner phase references
- **Spectrum Efficiency:** PM can have more compact spectrum in some cases